Technology Stack

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	2 Marks

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	Frontend / Client-Side	HTML5, custom CSS, vanilla JavaScript, Chart.js	No framework overhead needed for 3 pages; Chart.js gives fast, license-free charts for gauges/trends/donuts.
2	Backend / Server-Side	Python 3, Flask	Lightweight, well-documented, integrates directly with scikit-learn for serving the trained model.
3	Machine Learning	scikit-learn (Linear Regression, Random Forest)	Both easy to compare on R^2/MAE; Linear Regression won after log-transforming GNI to match the real HDI formula.
4	Data Handling	pandas, NumPy	Standard for cleaning UNDP CSV data and feature engineering.
5	Data Visualization	Matplotlib, Seaborn	Used offline to generate the 4 EDA figures shown on Home.
6	Database / Storage	SQLite (predictions.db)	Zero-config, file-based persistence; sufficient for a single-user demo, swappable for Postgres later.
7	Model Serialization	joblib	Standard, fast serialization for scikit-learn models.
8	Deployment / Hosting	Render (Gunicorn WSGI server)	Free tier, native GitHub integration, simple build/start command configuration.
9	Version Control	Git + GitHub	Industry-standard source control and public code hosting.